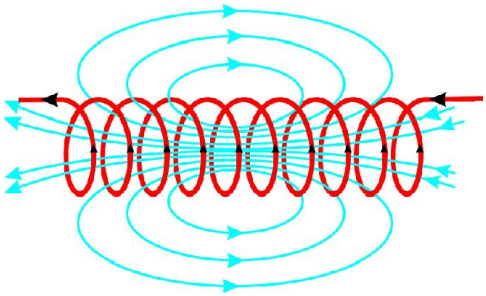
CLASS X · PHYSICAL SCIENCE · CHAPTER 10

Magnetic Effects of Electric Current

Student Workbook

Practise the way the exam asks.

Every topic of the chapter, turned into questions you answer in this book: fill in the blanks, label and draw field lines, apply the hand rules, complete the derivation, solve the numerical, explain the reason. Each unit ends with a self check, and the book ends with a full 40-mark chapter test.



- Marks badges: 1M 2M 4M
- Academic standards: AS1 concept AS3 experiment AS5 diagram AS7 application
- Textbook marks every question from *Improve your learning*

Name _____

Class / Section _____

Roll No. _____

School _____

How to use this workbook

- Read the **Recall in 60 seconds** box first. If a point feels new, go back to the textbook page before you start.
- Answer in the book, in pen. Do the questions **without** looking at the textbook, then check.
- Numericals: *formula → values in SI units (m, m^2, A, T) → answer with unit.*
- Field-line diagrams: draw in pencil, put **arrows on every line**, mark N and S, ⊙ and ⊗.
- For every direction question, write which **rule** you used (thumb rule, right hand rule, Lenz's law).
- Fill in the **Self check** after each unit; copy mistakes into the **Mistake log**.

What the exam asks from this chapter

Marks	Typical questions
1 mark	Units (weber, tesla), one-line laws, energy change in motor/generator, force when parallel to B, MCQs.
2 marks	Direction of current/field/force from a figure, pole of a coil, TV distortion, closed field lines, short numericals on $B = \Phi/A$, $F = ILB$, $\varepsilon = Blv$, $\varepsilon = N\Delta\Phi/\Delta t$.
4 marks	Activities: current produces a field; force on a wire; Faraday's law. Working of motor and generators. Derivation of $\varepsilon = Blv$, $F = ILB$, $r = mv/Bq$. Lenz's law with energy argument. Levitation.
Diagrams	Field lines of bar magnet, straight wire, solenoid; right hand rules; coil in a field; motor; Faraday's activity; AC generator and emf graph.

My progress tracker

Unit	Topic	Score	Confidence	Revised on
1	Oersted, magnetic field and field lines	_____	☐☐☐	_____
2	Magnetic flux and flux density	_____	☐☐☐	_____
3	Fields of a wire, coil and solenoid	_____	☐☐☐	_____
4	Force on a moving charge	_____	☐☐☐	_____
5	Force on a current; electric motor	_____	☐☐☐	_____
6	Induction: Faraday, Lenz, motional emf	_____	☐☐☐	_____
7	Applications, levitation, generators	_____	☐☐☐	_____
–	One-mark rapid fire	_____	☐☐☐	_____
–	Chapter test (40 marks)	_____	☐☐☐	_____

Confidence: tick one box for "shaky", two for "fine", three for "I can teach it".

Name: _____

UNIT 1

Magnetic field and field lines

1M 2M 4M

Recall in 60 seconds

- **Oersted (1820)**: a compass needle under a wire deflects when current flows ⇒ a current produces a magnetic field.
- A **magnetic field** is the region around a magnet where its force acts without contact. It has **strength** and **direction** and is three-dimensional.
- **Field lines** are imaginary; the **tangent** gives the direction; crowded lines = strong field. Outside a magnet they go N → S. Uniform field: parallel, equally spaced lines.

A Fill in the blanks

AS1 1M

1. The scientist who first showed that a current deflects a compass needle was _____.
2. The direction of the field at a point is given by the _____ to the field line at that point.
3. Field lines are _____ near the poles, where the field is _____.
4. A field whose strength and direction are the same everywhere is called a _____ field.
5. Outside a bar magnet field lines go from _____ to _____.

B True or false

AS1 1M

Write T or F. If false, correct it on the line below.

1. Field lines of a magnet can cross each other. [ ]

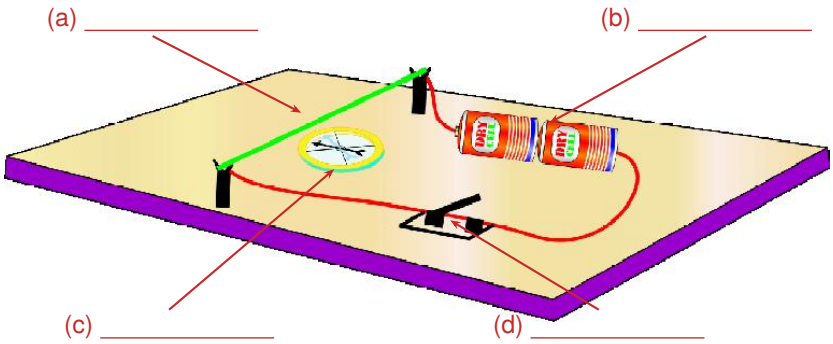
2. The magnetic field of a bar magnet exists only in the plane of the paper. [ ]

3. Far from a magnet a compass needle points almost north–south. [ ]

C Label the Oersted apparatus

AS5 2M

Name: _____



1. What happens to the needle when the key is closed? What does it prove? 2M

D Draw field lines

AS5 2M

Draw at least six field lines of the bar magnet below, with arrows. 2M



E Short answers

Textbook AS1 2M

1. Are magnetic field lines closed? Explain. 2M

Name: _____

2. How can you find the strength and the direction of a magnetic field using field lines? **2M**

Self check

I can . . .

- ☐ describe Oersted's experiment and its conclusion
- ☐ state the properties of field lines
- ☐ draw the field lines of a bar magnet with arrows

My score

_____ / 19

Redo every question you got wrong on the **Mistake log** page.

Name: _____

UNIT 2

Magnetic flux and flux density

1M 2M

Recall in 60 seconds

- **Magnetic flux** Φ: number of field lines through a plane held perpendicular to the field. Unit **weber (Wb)**.
- **Flux density** $B = \dfrac{\Phi}{A}$ (area perpendicular to the field). Unit **Wb/m^2 = tesla (T)**.
- Plane tilted so its normal makes θ with B: effective area $A\cos\theta$, so $\Phi = BA\cos\theta$. Plane parallel to B: $\Phi = 0$.

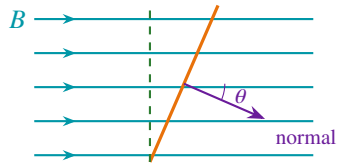
A Fill in the blanks

AS1 1M

1. The SI unit of magnetic flux is the _____.
2. The SI unit of magnetic flux density is the _____, which equals _____.
3. Magnetic flux density is also called magnetic _____.
4. The flux through a plane held parallel to the field is _____.

B Complete the derivation of $\Phi = BA\cos\theta$

AS1 2M



edge view of a plane of area A

(1) B uses the area _____ to the field:

$$B = \frac{\Phi}{A_\perp}$$

(2) The effective area perpendicular to the field is $A_\perp$ = _____.

(3) So B = _____ and Φ = _____.

(4) $\theta = 0°$: Φ = _____; $\theta = 90°$: Φ = _____.

C Numericals

AS7 2M

Worked example: 0.02 Wb crosses 0.4 m^2 normally

$B = \dfrac{\Phi}{A} = \dfrac{0.02}{0.4} = 0.05$ T. **Always** convert cm^2 to m^2 (1 cm^2 = 10^{-4} m^2).

1. $B = 0.8$ T passes normally through a loop of area 0.05 m^2. Find the flux. 2M

2. The same loop is turned so that its normal makes 60° with the field. Find the new flux. 2M

Name: _____

3. A flux of 6×10^{-4} Wb passes normally through 20 cm^2. Find B. **2M**

Self check

I can . . .

- ☐ define magnetic flux and flux density with units
- ☐ derive $\Phi = BA\cos\theta$ and use its special cases
- ☐ convert areas to m^2 in numericals

My score

_____ / 15

Redo every question you got wrong on the **Mistake log** page.

Name: _____

UNIT 3

Fields of a wire, a coil and a solenoid

1M 2M 4M

Recall in 60 seconds

- **Straight wire**: field lines are concentric circles. **Right hand thumb rule**: thumb along current, curled fingers give the field. Out of the page ⊙ ⇒ anticlockwise; into the page ⊗ ⇒ clockwise.
- **Circular coil**: field at the centre is perpendicular to the plane of the coil. Curl fingers along the current, thumb gives B. Anticlockwise current (as seen) ⇒ that face is **N**.
- **Solenoid**: behaves like a bar magnet; outside N → S, inside S → N; lines are closed loops.

A Draw the field lines

AS5 2M

Draw three field lines with arrows around each wire. 2M



(1) current out of the page



(2) current into the page

B Direction questions from the textbook

Textbook AS1 1M 2M

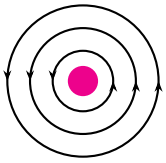


fig-Q1

1. Field lines around a wire are shown in fig-Q1 (anticlockwise). What is the direction of the current? 1M

Name: _____

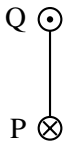
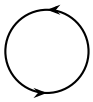


fig-Q3

2. A coil is kept perpendicular to the page. At P the current flows into the page, at Q it comes out (fig-Q3). What is the direction of the field due to the coil? 2M

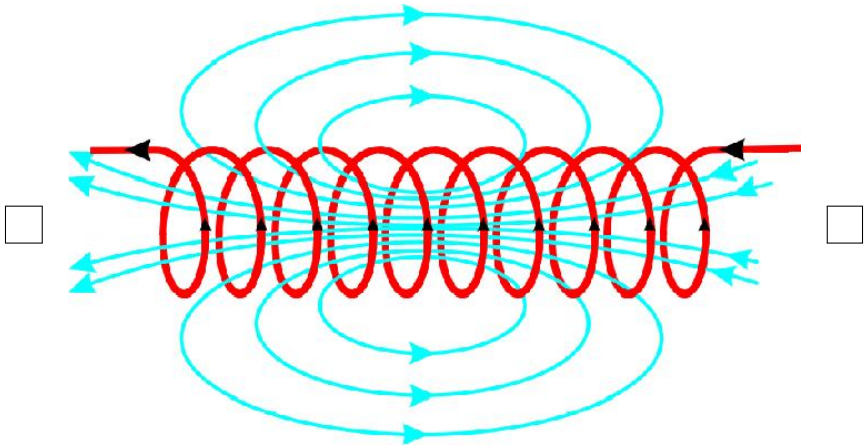

fig-Q4

3. The current in a coil flows as in fig-Q4 (anticlockwise as you see it). What pole is formed at this face? 1M

C Label the solenoid

AS5 2M

In the figure the current enters at the right end and leaves at the left. Mark the N and S ends and the direction of the field inside.



D Experiment

Textbook AS3 4M

Name: _____

Explain with the help of two activities that a current-carrying wire produces a magnetic field. 4M

Self check

I can . . .

- ☐ draw the field of a straight wire for both current directions
- ☐ find the pole of a coil face from the current direction
- ☐ describe the field of a solenoid and why its lines are closed loops

My score

_____ / 20

Redo every question you got wrong on the **Mistake log** page.

Name: _____

UNIT 4

Force on a moving charge

1M 2M 4M

Recall in 60 seconds

- A magnet distorts the picture on a CRT TV: the field exerts a force on the moving electrons.
- $F = qvB\sin\theta$; $F = qvB$ when $v \perp B$; $F = 0$ when $v \parallel B$ or $v = 0$.
- Direction: right hand rule (forefinger v, middle finger B, thumb F), for a **positive** charge; reverse it for an electron. F is always perpendicular to both v and B.
- A charge moving $\perp B$ goes round a circle: $r = \dfrac{mv}{Bq}$, $T = \dfrac{2\pi m}{Bq}$.

A Fill in the blanks

AS1 1M

1. The force on a charge moving parallel to the field is _____.
2. The magnetic force is always perpendicular to both the _____ and the _____.
3. For an electron, the direction of force is _____ to that found for a positive charge.
4. The magnetic force on a charge moving $\perp B$ acts as the _____ force.

B Complete the derivation

AS1 4M

A charge q of mass m moves with speed v perpendicular to a uniform field B.

(1) Magnetic force: F = _____ (since $\sin 90° =$ _____).

(2) This force is always perpendicular to the velocity, so the path is a _____.

(3) Centripetal force = magnetic force: _____ $= qvB \Rightarrow r =$ _____

(4) Time period $T = \dfrac{2\pi r}{v}$ = _____, which does _____ depend on the speed.

C Numericals

AS7 2M

1. A charge of 2×10^{-6} C moves at 3×10^{4} m/s perpendicular to $B = 0.5$ T. Find the force. 2M

2. The same charge moves at 30° to the field. Find the force. 2M

3. A proton ($m = 1.67 \times 10^{-27}$ kg, $q = 1.6 \times 10^{-19}$ C) moves at 10^{6} m/s perpendicular to $B = 1$ T. Find the radius

Name: _____

of its path. 2M

D Short answers

Textbook AS1 2M

1. Why does the picture appear distorted when a bar magnet is brought close to the screen of a television? 2M

2. State the right hand rule for the direction of force on a positive charge. 2M

Self check

I can . . .

- ☐ use $F = qvB\sin\theta$ including the zero-force cases
- ☐ find the direction of force with the right hand rule, and reverse it for electrons
- ☐ derive $r = mv/Bq$ and $T = 2\pi m/Bq$

My score

_____ / 25

Redo every question you got wrong on the **Mistake log** page.

Name: _____

UNIT 5

Force on a current; electric motor

1M 2M 4M

Recall in 60 seconds

- A current is moving charge, so a current-carrying wire in a field feels a force: $F = ILB\sin\theta$; $F = ILB$ when $I \perp B$; $F = 0$ when $I \parallel B$.
- Activity 8: a wire in a horseshoe magnet is deflected; reversing the current or the poles reverses the deflection.
- **Motor**: forces on opposite sides of a coil are equal and opposite (net force 0) but form a **couple** that rotates the coil. **Split rings** and brushes reverse the current every half turn. Electrical → mechanical energy.

A Complete the derivation of $F = ILB$

AS1 4M

(1) Force on one charge moving with drift velocity v perpendicular to B: F_0 = _____.

(2) Total charge in the length L inside the field is Q: F = _____ . . . (1)

(3) Time to cross length L: $t = \frac{L}{v}$, so v = _____ . . . (2)

(4) Put (2) in (1): F = _____ . . . (3)

(5) Since $\frac{Q}{t}$ = _____, F = _____. At an angle θ: F = _____.

B Numericals

Textbook AS7 2M

1. Textbook An 8 N force acts on a 20 cm conductor placed perpendicular to a magnetic field. The current is 40 A. Find the magnetic field induction. 2M

2. A wire 0.5 m long carries 4 A at 30° to a field of 0.2 T. Find the force. 2M

C Direction of force

Textbook AS1 2M

Name: _____

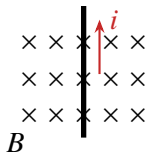
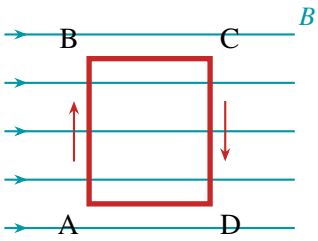


fig-Q5

A straight wire carrying current i upward is kept perpendicular to a field directed into the page; length L is in the field. Find the magnitude and direction of the force. 2M

D Motor

AS1 AS5 4M



1. The force on AB is directed _____ and on CD _____. 1M
2. The net force on the coil is _____, yet the coil _____ because the two forces form a _____. 1M
3. Why is the current reversed every half turn, and which part does it? 2M

E Experiment

Textbook AS3 4M

How do you verify experimentally that a current-carrying conductor experiences a force when kept in a magnetic

Name: _____

field? 4M

Self check

I can . . .

- ☐ derive $F = ILB$ from $F = qvB$
- ☐ find the direction of force on a wire with the right hand rule
- ☐ explain the working of a motor, including split rings

My score

_____ / 27

Redo every question you got wrong on the **Mistake log** page.

Name: _____

UNIT 6

Induction: Faraday, Lenz and motional emf

1M 2M 4M

Recall in 60 seconds

- **Electromagnetic induction**: a current is induced in a coil when the flux through it changes (relative motion of magnet and coil).
- **Faraday's law**: $\varepsilon = \dfrac{\Delta\Phi}{\Delta t}$; for N turns $\varepsilon = N\dfrac{\Delta\Phi_0}{\Delta t}$.
- **Lenz's law**: the induced current opposes the change in flux (conservation of energy). N pole approaching $\Rightarrow$ near face of coil becomes N.
- **Motional emf**: $\varepsilon = Blv$ for a conductor of length l moving with speed v perpendicular to B.

A Activity 9: what does the galvanometer show?

AS3 1M

Action	Galvanometer
Magnet held at rest inside the coil	_____
N pole pushed towards the coil	_____
N pole pulled away from the coil	_____
S pole pushed towards the coil	_____
Magnet pushed in faster	_____

B Fill in the blanks

AS1 1M

1. The current produced by changing flux is called the _____.
2. Lenz's law is a consequence of the conservation of _____.
3. When the N pole of a magnet approaches a coil, the face of the coil nearer to it becomes a _____ pole.
4. $\varepsilon = Blv$ is called the _____.

C Complete the derivation of $\varepsilon = Blv$

AS1 4M

A cross wire of length l lies on two parallel conductors in a uniform field B. It is moved a distance s in time Δt and a current I flows.

(1) Force of the field on the cross wire: $F =$ _____ . . . (8)

(2) Work done by us in moving it: $W = Fs =$ _____ . . . (9)

(3) Change in flux as the loop area changes by ls: $\Delta\Phi =$ _____ . . . (10) $\Rightarrow W =$ _____

(4) Dividing by Δt: $P = \dfrac{W}{\Delta t} =$ _____; comparing with $P = \varepsilon I$ gives $\varepsilon =$ _____

(5) Also $P = Fv =$ _____ (since $\dfrac{s}{\Delta t} = v$). So $\varepsilon I = BIlv \Rightarrow \varepsilon =$ _____

D Numericals

Textbook AS7 2M

Name: _____

Worked example (textbook): 400 turns, flux per turn rises by 0.001 Wb in 0.1 s

$\varepsilon = N\dfrac{\Delta\Phi_0}{\Delta t} = \dfrac{400 \times 0.001}{0.1} = 4$ V. While the flux stays constant (0.1 s to 0.3 s), $\varepsilon = 0$.

1. Textbook Find the length of a conductor moving at 10 m/s perpendicular to a field of 0.8 T if it induces an emf of 8 V. 2M

2. The flux per turn in a coil of 100 turns falls from 0.05 Wb to 0.01 Wb in 0.2 s. Find the induced emf. 2M

3. A 1.5 m rod moves at 2 m/s perpendicular to $B = 0.4$ T. Find the emf. 2M

E Figure questions

Textbook AS1 AS2 2M

1. A bar magnet with its N pole facing a coil moves towards it (fig-Q2). What happens to the flux through the coil? Which way does the induced current flow as seen from the magnet? 2M

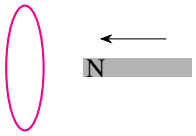


fig-Q2

Name: _____

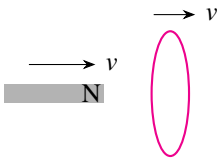


fig-Q6

2. Textbook The coil and the bar magnet both move in the same direction with the same speed (fig-Q6). What happens? 2M

F Long answers

Textbook AS1 AS3 4M

1. Explain Faraday's law of induction with the help of an activity. 4M

Name: _____

2. State Lenz's law and show that it follows from the conservation of energy. **4M**

3. What experiment do you suggest to understand Faraday's law? What instruments are required? Give suggestions for good results and precautions. **4M**

Name: _____

Self check

I can . . .

- ☐ describe Activity 9 and state Faraday's law
- ☐ state Lenz's law and give the energy argument
- ☐ derive $\varepsilon = \Delta\Phi/\Delta t$ and $\varepsilon = Blv$
- ☐ solve induced-emf numericals with N turns

My score

_____ / 44

Redo every question you got wrong on the **Mistake log** page.

Name: _____

UNIT 7

Applications, levitation and generators

1M 2M 4M

Recall in 60 seconds

- Applications of induction: security metal detectors, tape recorder heads, induction stoves, ATM card readers.
- **Levitation**: AC in a coil around an iron core ⇒ changing flux through a metal ring ⇒ induced current opposes it ⇒ repulsion balances the ring's weight. With DC the ring only jumps at switch on/off.
- **AC generator**: coil rotated in a field; full **slip rings**; emf zero when the coil is vertical, maximum when horizontal; reverses every half turn. **DC generator**: half (split) rings keep the output in one direction. Mechanical → electrical energy.

A Match the following

AS1 1M

	Column A		Column B
1.	Induction stove	[ ]	a) Induced current in a large coil triggers an alarm
2.	Security check	[ ]	b) Magnetised tape moves past a small coil
3.	Tape recorder	[ ]	c) Induced current heats a metal pan
4.	Electric motor	[ ]	d) Mechanical energy into electrical energy
5.	Generator	[ ]	e) Electrical energy into mechanical energy

B Levitation

AS1 AS7 2M 4M

1. Why does the metal ring levitate when AC is used? 4M

Name: _____

2. What happens if DC is used instead of AC? Why? 2M

C AC generator

AS1 AS5 2M 4M

1. Sketch the emf produced by an AC generator over one rotation, starting with the coil vertical. Mark where the emf is zero and maximum. 2M

2. Why does the emf vary from zero to maximum during a rotation? 2M

3. How is an AC generator converted into a DC generator? 2M

D Textbook application

Textbook AS7 2M

Name: _____

Give a few applications of Faraday's law of induction in daily life. 2M

Self check

I can . . .

- ☐ explain four applications of induction
- ☐ explain levitation with AC and why DC fails
- ☐ explain the AC generator, its emf graph and the DC version

My score

_____ / 25

Redo every question you got wrong on the **Mistake log** page.

Name: _____

One-mark rapid fire

Answer in one word or one line. Target **8 minutes** for 25 questions. Time taken: _____ Score: _____ / 25

1. Discovered the magnetic effect of current? _____
2. Unit of magnetic flux? _____
3. Unit of flux density? _____
4. 1 T in N, A, m? _____
5. Φ for a plane $\parallel B$? _____
6. Field lines of a straight wire? _____
7. Current $\odot$: field lines are? _____
8. Rule for field of a wire? _____
9. Anticlockwise current face is? _____
10. Coil that acts like a bar magnet? _____
11. Force on a charge moving $\parallel B$? _____
12. Formula for force on a charge? _____
13. Radius of path of a charge? _____
14. Force on a current $\perp B$? _____
15. Motor converts? _____
16. Reverses current in a motor? _____
17. Faraday's law formula? _____
18. Law giving induced current direction? _____
19. Lenz's law is based on? _____
20. Motional emf? _____
21. Emf when coil plane $\perp B$ (generator)? _____
22. Rings in an AC generator? _____
23. Generator converts? _____
24. Stove using induction? _____
25. Current reversing periodically? _____

Name: _____

Score boosters

Anatomy of a full-marks 4-mark answer

Example question: *Explain the working of an electric motor.*

Part	What to write	Why it matters
1. Principle	A current-carrying coil in a field experiences a force ($F = ILB$).	Scores the concept mark.
2. Diagram	Coil ABCD between N and S, split rings, brushes, battery; arrows for current.	Diagrams carry marks.
3. Forces	AB and CD: equal, opposite forces $\Rightarrow$ couple $\Rightarrow$ rotation.	Shows you know why.
4. Commutator	Split rings reverse current every half turn $\Rightarrow$ continuous rotation.	Key idea examiners look for.
5. Energy	Electrical energy $\rightarrow$ mechanical energy.	Closes the answer.

Keywords the examiner looks for

Oersted	compass deflects · current produces field
Field lines	tangent gives direction · crowded = strong · closed loops
Straight wire	concentric circles · right hand thumb rule
Force	$qvB\sin\theta$ · $ILB\sin\theta$ · zero when parallel · right hand rule
Motor	couple · split rings · brushes · electrical to mechanical
Faraday	change of flux · rate of change · induced emf
Lenz	opposes the change · conservation of energy
Generator	rotating coil · slip rings · zero/maximum emf · mechanical to electrical

Mistakes that cost marks

- Field lines without arrows.
- θ measured from the plane, not the normal, in $\Phi = BA\cos\theta$.
- Forgetting $F = 0$ when parallel to B.
- Not reversing the force for an electron.
- Using cm or cm^2 without converting.
- Forgetting N in $\varepsilon = N\Delta\Phi_0/\Delta t$.
- Slip rings in a motor (it has split rings).
- No energy argument in Lenz's law.

Name: _____

Formula sheet (cover the right column and test yourself)

Quantity	Formula / unit
Flux density	$B = \dfrac{\Phi}{A}$ tesla = Wb/m^2
Flux	$\Phi = BA\cos\theta$ weber
Force on a charge	$F = qvB\sin\theta$
Circular motion	$r = \dfrac{mv}{Bq}$, $T = \dfrac{2\pi m}{Bq}$
Force on a current	$F = ILB\sin\theta$
Faraday's law	$\varepsilon = \dfrac{\Delta\Phi}{\Delta t}$; $\varepsilon = N\dfrac{\Delta\Phi_0}{\Delta t}$
Motional emf	$\varepsilon = Blv$

Diagram practice tracker

Draw each diagram three times on separate days. Tick when you can draw it from memory in under 4 minutes.

Diagram	Try 1	Try 2	Try 3
Field lines of a bar magnet	☐	☐	☐
Field of a straight wire (⊙ and ⊗)	☐	☐	☐
Field of a solenoid with N and S	☐	☐	☐
Right hand rule for force	☐	☐	☐
Electric motor with split rings	☐	☐	☐
Faraday's activity (magnet, coil, galvanometer)	☐	☐	☐
AC generator and emf graph	☐	☐	☐

Name: _____

Chapter test

Time: 1 hour 30 minutes **Maximum marks: 40**

Section I: 6 questions × 1 mark. Section II: 5 questions × 2 marks. Section III: 4 questions × 4 marks, each with an internal choice. Section IV: 8 multiple choice questions × 1 mark.

Section I

6 × 1 = 6

1. What is the SI unit of magnetic flux density?

2. State Lenz's law.

3. What is the force on a current-carrying wire kept parallel to a magnetic field?

4. Which device converts mechanical energy into electrical energy?

5. Write the formula for motional emf.

6. Name the rule used to find the direction of the field around a straight wire.

Section II

5 × 2 = 10

7. Why does a TV picture get distorted when a magnet is brought near it?

Name: _____

8. A 0.5 m wire carries 2 A perpendicular to a field of 0.4 T. Find the force.

9. A coil of 50 turns has its flux per turn changed by 0.02 Wb in 0.1 s. Find the induced emf.

10. Are magnetic field lines closed? Explain.

11. A coil and a magnet move together in the same direction at the same speed. Is a current induced? Why?

Section III

$4 \times 4 = 16$

12. Explain with two activities that a current-carrying wire produces a magnetic field.

Name: _____

OR Describe an activity to show that a current-carrying wire experiences a force in a magnetic field.

13. Derive $F = ILB$ for a straight wire in a magnetic field.

OR Derive $r = \frac{mv}{Bq}$ and $T = \frac{2\pi m}{Bq}$ for a charge moving perpendicular to a field.

14. Derive $\varepsilon = Blv$ using conservation of energy.

Name: _____

OR State Faraday's law and explain it with an activity.

15. Explain the working of an electric motor with a diagram.
OR Explain the working of an AC generator with the graph of emf for one rotation.

Section IV

$8 \times 1 = 8$

Name: _____

16. Which converts electrical energy into mechanical energy? [ ]

A) motor B) battery

C) generator D) switch

17. The force on a wire perpendicular to a uniform field is [ ]

A) 0 B) ILB

C) $2ILB$ D) $ILB/2$

18. One tesla is equal to [ ]

A) N/C B) N/(A m)

C) A/m D) N/(A s)

19. Unit of magnetic flux is [ ]

A) dyne B) oersted

C) gauss D) weber

20. No force acts on a current-carrying conductor kept [ ]

A) parallel to the field B) perpendicular to the field

C) in the field D) away from the field

21. Current flows anticlockwise in a coil as you see it. The face towards you is [ ]

A) S pole B) N pole

C) neither D) both

22. In an AC generator the emf is maximum when the plane of the coil is [ ]

A) perpendicular to B B) parallel to B

C) at 45° D) at rest

23. The time period of a charge moving in a circle in a uniform field depends on [ ]

A) its speed B) the radius

C) m, q and B D) none

My test result

Section	I	II	III	IV	**Total**
Marks	_____/6	_____/10	_____/16	_____/8	_____/40

Name: _____

Mistake log

Write every question you got wrong. Redo it after one week without looking at the answer.

Unit / Q	What I wrote	The correct idea	Redone
			☐
			☐
			☐
			☐
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			☐
			☐
			☐
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Name: _____